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Studierichting: Informatica
Oefeningenreeks 1C nr. 46.2

$z^4 = -1$

$r = |z| = \sqrt{(-1)^2 + 0^2} = \sqrt{1} = 1$

$\tan(\theta) = \dfrac{0}{-1} = 0$

$\theta = 0$

We tellen hierbij π op want onze a is negatief:

$\Rightarrow \theta = \pi$

$\Rightarrow z = 1 \cdot \text{cis}(\pi) = \text{cis}(\pi)$

$$w_0 = \text{cis}\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) + i \cdot \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2} \cdot i}{2}$$
$$w_1 = \text{cis}\left(\frac{\pi + 2\pi}{4}\right) = \cos\left(\frac{3\pi}{4}\right) + i \cdot \sin\left(\frac{3\pi}{4}\right) = \frac{-\sqrt{2}}{2} + i \cdot \frac{\sqrt{2}}{2}$$
$$w_2 = \text{cis}\left(\frac{\pi + 4\pi}{4}\right) = \text{cis}\left(\frac{5\pi}{4}\right) = \cos\left(\frac{5\pi}{4}\right) + i \cdot \sin\left(\frac{5\pi}{4}\right) = \frac{-\sqrt{2}}{2} - \frac{\sqrt{2} \cdot i}{2}$$
$$w_3 = \text{cis}\left(\frac{\pi + 6\pi}{4}\right) = \text{cis}\left(\frac{7\pi}{4}\right) = \cos\left(\frac{7\pi}{4}\right) + i \cdot \sin\left(\frac{7\pi}{4}\right) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2} \cdot i}{2}$$

References